

RESEARCH INTERESTS

Federated Learning, Optimization, Statistical Inference, Natural Language Processing, High-Dimensional Statistics, Networks, Reinforcement Learning

EDUCATION

North Carolina State University

Ph.D. in Statistics (Advisor: Dr. Srijan Sengupta)

Expected Graduation Date: May 2027

Raleigh, NC, USA

Aug. 2022 - Present

Indian Statistical Institute, Kolkata

Master of Statistics

Kolkata, WB, India

2020 - 2022

Indian Statistical Institute, Kolkata

Bachelor of Statistics

Kolkata, WB, India

2017 - 2020

CONFERENCE PAPERS

Understanding Narratives of Trauma in Social Media

2025

*Mansi Saxena, Vaibhav Garg, **Bhaskar Ray**, Aura Mishra, Munindar Singh*

Accepted at the 17th ACM Web Science Conference 2025

PH.D. DISSERTATION PROJECTS

PACE: Privacy Aware Collaborative Estimation for Heterogenous GLMs

Jan. 2023 - Present

Department of Statistics, North Carolina State University

Supervisors: Dr. Srijan Sengupta, Dr. Aritra Mitra

We propose a novel two-step method of parameter estimation for a broad class of Generalized Linear Models under a distributed data framework with privacy restrictions on data sharing. Participating sites obtain data from the same distribution, the true parameter values varying across clusters. We conduct a hypothesis test-type clustering and subsequently, a decentralized collaborative estimation and compare the convergence and accuracy guarantees to local estimation. R package `sgd` was modified as `sgditer` (GitHub) for implementation of proposed method.

On Federated Learning with Robust Estimators in GLMs

Jan. 2025 - Present

Department of Statistics, North Carolina State University

Supervisors: Dr. Srijan Sengupta

We analyze the theoretical properties of traditional and modified robust parameter estimators applied to GLMs in a federated learning framework.

OTHER PROJECTS

A gut-brain interaction controlling reward learning

Aug. 2023 - Present

Department of Statistics, North Carolina State University

Supervisors: Dr. William Matthew Howe, Dr. Srijan Sengupta

In collaboration with Virginia Polytechnic Institute and State University, we are working on the statistical analysis of the timescale over which gut-derived signals based on food content modulate dopamine release, and how they interact with dopamine signals previously identified as critical for food reward learning and motivation.

Predicting Organic Crystal Solubilities using Machine Learning

Aug. 2022 - July 2023

Department of Statistics, North Carolina State University

Supervisors: Dr. Jacqueline M. Hughes-Oliver, Dr. Erik E. Santiso

In collaboration with the Department of Chemical and Biomolecular Engineering, we developed a predictive machine learning model for solubilities of different organic solute and solvent combinations to aid thermodynamic models built for predictions at extreme temperature and pressure.

On Ensembles for Missing Data using Modified Neural Networks

July 2021 - Present

Indian Statistical Institute, Kolkata

Supervisor: Dr. Smarajit Bose (Interdisciplinary Statistical Research Unit)

We explore conformal prediction in the light of modifications to ensemble learning methods exploring various methodologies to model missing data using neural networks without requiring imputations.

SELECTED CONFERENCES AND WORKSHOPS

2026 Frontiers in Learning Under Data Heterogeneity (*Upcoming*)

Poster on PACE: Privacy Aware Collaborative Estimation for Heterogenous GLMs

Jan. 16-17, 2026

Gainesville, FL, USA

2025 INFORMS Annual Meeting

Contributed talk on PACE: Privacy Aware Collaborative Estimation for Heterogenous GLMs

Oct. 26-29, 2025

Atlanta, GA, USA

Joint Statistical Meetings 2025

Contributed talk on PACE: Privacy Aware Collaborative Estimation for Heterogenous GLMs

Aug. 2-7, 2025

Nashville, TN, USA

IISA Conference 2024

Poster on Federated Learning with GLMs under Parameter Heterogeneity

Dec. 27-31, 2024

Kochi, Kerala, India

SRCOS Summer Research Conference

Poster on Federated Learning with GLMs under Parameter Heterogeneity

June 3-5, 2024

Clemson, SC, USA

EXPERIENCE

Graduate Research Assistant (Data Science Research Consultant)

Department of Data Science and AI Academy, NCSU

Spring 2026

Graduate Research Assistant

Department of Statistics, NCSU (Supervisor: Dr. Srijan Sengupta)

Spring - Fall '25, Fall '23 - Spring '24

Graduate Teaching Assistant

Department of Statistics, NCSU

Fall 2024

- Applied Time Series (Supervisor: Dr. Donald E. K. Martin)
- Statistical Methods I (Supervisor: Dr. Jonathan Stallrich)

Graduate Research Assistant

Department of Statistics, NCSU (Supervisor: Dr. Srijan Sengupta)

July 2023 - Aug. 2023

Graduate Research Assistant

Department of Statistics, NCSU (Supervisor: Dr. Jacqueline Hughes-Oliver)

Fall 2022 - Summer 2023

AWARDS AND ACHIEVEMENTS

Travel Award for 2026 Frontiers in Learning Under Data Heterogeneity

January 2026

SRC Conference Travel Award

June 2024

KVPY Fellowship Award

2017 - 2022

Funded by the Department of Science and Technology, Govt. of India. Selected in 2015.

Regional Mathematics Olympiad

2015

West Bengal, India

LEADERSHIP ROLES

Treasurer, Triangle Bengali Association

North Carolina State University

Spring 2023 - Spring 2025